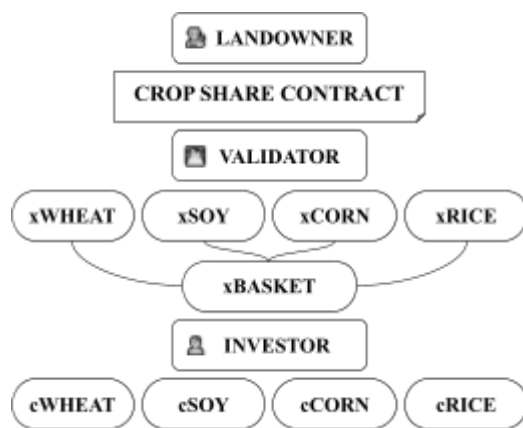


LandX - The Perpetual Commodity Vaults Protocol

ABSTRACT. Perpetual commodity vaults provide investors an inflation hedged return backed by a legal contract secured on underlying farmland. LandX makes perpetual vaults available as a liquid digital asset.

INTRODUCTION

LandX offers tokenized commodities and perpetual vaults for agricultural products.



cTokens are on-chain commodities representing 1^{KG} of the underlying product

xTokens are perpetual commodity vaults paying out 1 cToken per year in yield

The xToken perpetual yield is derived from 49 year rolling legal crop share contracts called liens, secured by farmland and managed by validators.

An internal marketplace offers cToken/USDC liquidity enabling users to trade commodities without slippage at an oracle set rate.

At launch there are four markets representing the most important and widely traded agricultural commodities: Wheat, Soy, Corn & Rice.

xBASKET is an ERC20 token which represents a basket of equally weighted xTokens. This provides further diversification offering exposure to all four agricultural commodities in a single *index fund token*.

Users can at any time mint or redeem their xBASKET for the individual underlying xTokens.

“LandX provides the opportunity to generate real yield, from real production and bring that on-chain to create sustainable DeFi incentives”

cTokens & xTokens are liquid digital assets traded 24/7. cTokens value derives from the market value of the commodity. xToken value derives from investors pricing of the vault. If the value of wheat goes up then the amount of yield paid per year per xToken increases.

In traditional finance agricultural products account for 5% of derivative trading volumes. LandX has a first mover advantage in DeFi.

LandX provides landowners with the upfront capital required to grow their businesses, invest in equipment and modernize their production capabilities. It is more flexible, efficient and fair than the traditional financing that is currently available.

For investors LandX provides exposure to an uncorrelated digital asset which acts as an inflation hedge and brings real yield to DeFi markets.

xTOKENS DEFI DIVERSIFICATION

Crypto-native investment portfolios are highly correlated to the cryptocurrency and stock markets. There is currently a lack of products within the DeFi ecosystem to gain exposure outside of these two core markets.

The xToken perpetual vaults are the primary investment product which offer the following:

- **Portfolio diversification** through exposure to uncorrelated assets. Agricultural commodities, in the past, have moved independently of the stock market and crypto market.
- **Inflation hedge** because yield payments are made in cTokens which track the value of the underlying commodity. If the value of the USD falls over time relative to food products then the yield increases in USD terms due to the payment being calculated on 1^{KG} of the product.

Landowners are required to keep 12 months of crop share as a security deposit and are always required to pre-pay their crop share obligations in advance.

Commodity yield is distributed to xToken stakers in cTokens that can be exchanged via the platform at a fair market value of 1KG of the underlying product.

1 staked xWHEAT token will earn 1 cWHEAT per year which can be traded for the market value of 1^{KG} of wheat

xTokens and the xBASKET index token are tradeable in well backed liquidity pools on Uniswap v3, the leading decentralized exchange.

cTOKENS COMMODITY COLLATERAL

cTokens are derivatives designed to track the price of an agricultural asset.

This opens up new possibilities for DeFi protocols to use a new type of digital asset for collateral.

Currently DeFi protocols are either built around the native asset ETH or various USD stablecoins. cToken commodities offer reduced volatility without the long term devaluation of fiat based stablecoins.

cTokens are tradeable via an internal marketplace which collects landowner yield in USDC and offers it at market value for cTokens.

cToken exchange prices are updated via a 3rd party oracle which uses futures pricing data from traditional finance exchanges.

All tokens are ERC20 compatible and deployed to the Ethereum mainnet.

In due course the assets will be bridged across to popular layer 2 rollups with market leaders including Optimism and Arbitrum currently being considered.

LANDX OPERATIONS

There are three LandX ecosystem participants

Landowners commit their farmland crop share to LandX to access capital from investors. Their commitment is secured by a lien on the underlying farmland.

Validators are the bridge between farmers and the protocol. They're responsible for onboarding new farmland and assisting with the process's legal and economic contracts.

Investors deploy capital in exchange for exposure to real farmland assets through xToken perpetual commodity vaults.

The process begins with a landowner that wishes to raise capital. LandX provides farmers with capital in exchange for a share of their future crop. These legal contracts are secured by a lien over the farmer's land.

These contracts generate a permanent yield in return for an investment. The lien is managed by a validator who provides local legal expertise, and ensures timely crop share payments.

Validators connect landowners across the globe to the capital markets on LandX and are responsible for:

- Validating & creating a valuation for farmland
- Assisting with legal and financial contracts
- Staking LNDX tokens to provide investor protections against defaults

In return for their work, LandX validators receive a 1%¹ commission of the financing amount for bringing on farmers and to cover legal and administrative expenses.

Landowners receive xTokens equivalent to the crop-share they agree to.

The LandX platform requires farmers to keep 12 months of crop share payments on the platform in order to hedge against potential crop or cash uncertainties. Additionally, LandX requires a 12 month security deposit from every farmer.

Yield guaranteed by the landowners' contract is distributed from this pool of capital to xToken holders who can claim their commodity yield in USDC at any time.

In the event of default, the lien enables LandX to make a legal claim to the land and replace the non-paying farmer.

The lien contract and farmland collateral are represented on-chain as an NFT, which records the details of the crop share deal: area, location, crop share, title documents and lien. This NFT is available for viewing on the platform.

To receive xTokens the farmer deposits the NFT into the smart contract and receives an equivalent amount of xTokens.

If a farmer wants to exit their crop-share obligation, they can buy back the xTokens at current market value. Once the xTokens are returned to the smart contract the commitment becomes void.

LandX provides a more flexible financing route compared to what is currently available to landowners. At any time they can pay off their liability by buying back the xTokens or raise capital when required by selling their xTokens on a free market.

¹ Commission percentage subject to change via DAO vote

LANDX FOR LANDOWNERS

LandX provides funding based on the market value and production levels of the farmland used as collateral. By tapping into a broader, more sophisticated investor base LandX creates higher interest in agricultural derivative products which translates to higher valuations for the landowners.

The funding is structured so that the payments by the farmers are denominated in a fixed amount of crop share over a perpetual rolling 49-year period. This minimizes the financial impact on the landowner each year, ensuring the payments are fixed at the level that cannot exceed the MACS (Maximum Allowable Crop Share). This financing structure removes currency or commodity price risks.

Step 1. The landowner provides the validator with production yield data for five years, the cadastral records of the property, its geo coordinates and ownership information.

Step 2. Validator verifies the data, models the crop share and payout and presents the expected financing terms in USD;

Step 3. Property titles are pledged against the obligation to pay crop-share for rolling 49-year periods. A legal contract known as a lien is registered with local authorities.

Step 4. An NFT is minted with the terms of the contract.

Step 5. Farmers deposit the NFT into the smart contract to receive the xTokens. Farmers can sell the xTokens on the market or create an OTC deal.

Step 6. At any time the farmer can buy back the xTokens at market price on decentralized exchanges or OTC which will end the crop-share obligation.

Landowners can use the LandX platform to hedge against future yields while receiving capital and flexibility to invest in the tools needed to maximize the harvesting seasons.

For non-technical landowners, LandX provides support and an OTC desk to ensure everyone is capable of understanding and accessing the capital.

LandX improves upon the existing financial products available to farmers.

- There is no minimum amount of financing available. The amount of financing available is dependent on the productive area of the farmland pledged and its historical yield
- Farmers can repurchase the xTokens and burn them on the LandX protocol at any time to clear their commodity obligation
- LandX connects farmers to more sophisticated and efficient capital markets, increasing the valuation of their assets compared to traditional financing methods.

LandX provides farmers with the upfront capital required to grow their businesses, invest in equipment and modernize their production capabilities. It is more flexible, efficient and fair than the traditional financing that is currently available.

“For landowners LandX provides more flexible timely financing at better valuations with more options”

LANDX GOVERNANCE

LandX is structured as a decentralized autonomous organization (DAO) with voting rights belonging to holders of the LNDX governance token.

LNDX TOKEN MODEL

LandX offers something relatively unique in DeFi markets; real yield from real world production. This contrasts greatly with the majority of governance tokens which are built around zero-sum tokenomics.

LNDX value accrual ensures stakeholders interests are aligned with the project.

Validators are required to stake LNDX tokens as part of their commitment to the land agreements they set up.

Traders and investors in crypto markets seek out volatility and LNDX is the flagship way to gain exposure to LandX growth. While xTokens are expected to maintain strong correlations to their underlying assets, LNDX will likely fluctuate inline with crypto markets and the overall success of LandX.

LNDX token holders earn a percentage of platform fees from the xToken yield. These platform fees create utility and demand for LNDX due to the current and future value distribution. LNDX is priced on future earnings potential which makes it highly volatile and attractive as an altcoin investment.

Platform fees are set to 10% initially and distributed in USDC as follows:

- 60% LNDX token holders
- 35% LandX Treasury
- 5% LandX Choice

Fee percentage and distribution percentages are open to change in line with market conditions via a DAO proposal & vote. Valuation potential can be estimated using existing DeFi protocols and their TVL relative to their governance token market cap.

Protocol	TVL	Market Cap
Uniswap	\$5.0	\$5.1
Aave	\$5.5	\$1.3
MakerDAO	\$7.6	\$0.9
Curve	\$5.3	\$0.6

All values in Billions USD

Uniswap has a far greater market cap for its UNI governance token compared to competitors. This is partly due to the market leader position and also due to the fee structure and growing DAO treasury holdings of \$2.8B.

The LandX platform fee structure is set up to directly benefit the LNDX governance token holders. This makes the LNDX token very valuable from both a governance and fee distribution basis.

The LNDX token model incentivises collaboration and aligned interests of founders, investors, validators and stakeholders to work together and ensure the project is a success.

MARKET OPPORTUNITY

There are currently two target markets that would act as ideal customers for LandX:

1. DAO Treasuries looking to diversify crypto exposure to protect against market cycle downturns
2. Crypto-native investment funds looking for yield and inflation hedges

DAO TREASURIES

DAO treasuries have amassed huge amounts of capital with over \$10B USD now being managed by decentralized autonomous organizations. The top ten largest treasuries.

Name	Total Treasury	Liquid Assets
Uniswap	\$3,000,312,905	\$1,887,210,497
BitDAO	\$1,164,732,004	\$1,164,732,004
Lido	\$320,971,677	\$320,971,677
Olympus DAO	\$258,479,366	\$258,479,366
GnosisDAO	\$1,170,582,559	\$234,105,336
MakerDAO	\$155,431,818	\$155,431,818
Aave	\$154,021,768	\$154,021,768
ENS	\$731,418,552	\$137,627,283
Radicle	\$113,749,844	\$113,749,844
Aragon	\$98,990,983	\$98,990,983

Data from <https://openorgs.info>

LandX is looking for partners who wish to diversify their portfolios and DAO's are a perfect customer, they have a strong balance sheet that is overexposed to crypto assets.

CRYPTO NATIVE INVESTMENT FUNDS

There are many different types of investment funds in the crypto industry that have grown out of the sector's success. Many now manage billion dollar portfolios of digital assets such

as Alameda Research, Multicoins Capital, Paradigm & Wintermute. These funds often run a combination of strategies including:-

- Market Making
- Venture Capital
- Arbitrage Trading
- Basis Trading
- Directional Momentum

Funds are rarely delta-neutral entirely because of their exposure to low liquidity tokens. Their directional bias could be hedged or de-risked with an uncorrelated digital asset. Let's look at an example investment scenario.

A purchase of \$1,000,000 USD into the protocol's xBASKET index fund will give exposure to all four xTokens and return a yield expected to be in the region of 7% APR. Each xBASKET token returns the USDC value of 0.25 KGs of each commodity annually.

There is also expected appreciation of the commodity yield as the underlying commodities price increases. High inflation expectations suggest that commodities should increase in price relative to the USD over a long enough time period. This scenario creates a leveraged upside while at the same time providing a hedged downside for the investor.

If a basket of wheat, soy, corn and rice appreciates by 15% over the next year, and the investor sells their \$1M position, they'll profit approximately \$230,500 (23.05%). If the basket drops in price by 15% over the next year, they'll only realize a \$95K loss (-9.5%). Over longer term periods inflation further reduces the risk of short-term downside.

This is an asymmetric opportunity in an asset class only now opening up to treasury and fund managers. If the commodities go up, you get the increase in underlying and the yield. If the price goes down, some of your losses in the value of xTokens are offset by the daily yield received in USDC.

LIQUIDITY PROVISION

xTokens are traded primarily on Uniswap, the most popular decentralized exchange. Their pricing is dependent on supply, demand and liquidity provisions.

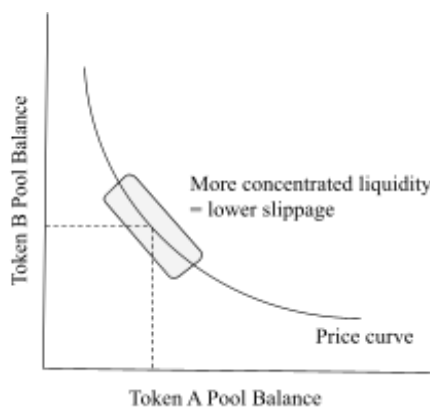
A LP (liquidity provider) will deposit a pair of assets to a liquidity pool in order to earn trading fees on that pair.

For example a LP can provide xSOY and USDC to the xSOY/USDC market on Uniswap. A trader can then come along and purchase the xSOY token with USDC (or another token which would be routed USD > USDC > xSOY).

This increases the balance of USDC in the xSOY/USDC pool and decreases the amount of available xSOY. This imbalance in the pool creates a shift in the price using the constant product formula.

$$x \times y = k$$

x and y represent the pool balance of each token, and k is the total constant price of the pool creating the following price curve.



Liquidity providers earn trading fees for providing this service at a rate of 0.3% per trade on standard pools.

Current commodity prices per KG as of July 2022 for price calculations.

- Wheat- \$0.35/kg
- Soy - \$0.57/kg
- Corn - \$0.31/kg
- Rice-\$0.42/kg
- xBasket - \$0.41/kg (*average*)

Initial prices can be calculated to target a 7% annual yield.

$$xTokenPrice = CommodityPrice \div 0.07$$

The LandX team will create the following pools initially and supply seed liquidity as described below.

Uniswap Pool	Initial Price	xToken	USDC
xBASKET/USDC	\$5.86	153,658	900,000
xWHEAT/USDC	\$5.00	200,000	1,000,000
xSOY/USDC	\$8.14	122,807	1,000,000
xCORN/USDC	\$4.43	158,064	700,000
xRICE/USDC	\$6.00	83,333	500,000

Exact amounts are subject to change based on commodity prices at time of deployment

Seed liquidity will be divided into two positions for each pool. 68% will put in a concentrated range to create a stable price. 32% will be put in a broad range to create wide liquidity in case of sudden spikes in demand².

Liquidity provisions will be managed on an ongoing basis with the aim of creating stable pricing relative to the underlying agricultural asset prices.

² Based on work by Guillaume Lambert
<https://lambert-guillaume.medium.com>

TECHNOLOGY

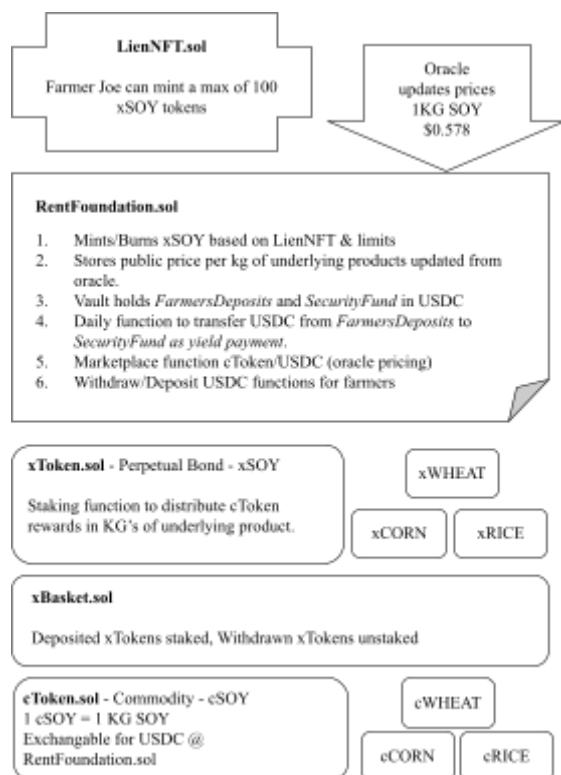
The LandX protocol is built on cutting edge blockchain technology. Blockchain is revolutionizing finance in the same way the internet disrupted commerce.

While cynics will point out bubbles, ponzis and failed projects the underlying technology is enabling a revolution in financial independence.

LandX smart contracts execute across a permissionless decentralized network. It has never been easier to connect an investor in Manhattan with a farmer in Pará.

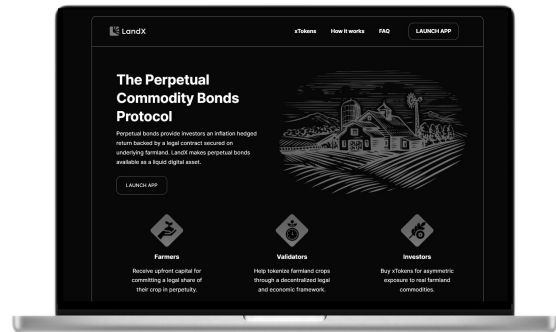
The benefits of this financial inclusion are still to play out but LandX is leading the way and excited to see how this technology breakthrough will change the world.

Smart Contracts



LandX Platform

The LandX platform is a dApp with a public web interface at: <https://LandX.fi>



The platform provides a central portal to buy, sell and manage xTokens.

Land Verification

LandX validators are responsible for identifying suitable farmland opportunities in their region, securing crop share agreements, as well as verifying the farmland productivity. LandX uses a combination of methods to verify the productivity of the land including the use of geospatial imagery. LandX evaluates *Normalized Difference Vegetation Index* (NDVI) for every block of land and compares the 5-year vegetation levels history with the data on regional averages and farmer reported actual crop yields.

LandX establishes a Maximum Allowable Crop Share (MACS) for every type of crop as a percentage of farmland yield that farmers are able to tokenize. MACS is one of the parameters that the LandX DAO will vote on. MACS is currently established at between 10-15% depending on the type of crop.

LandX validators are required to stake LNDX governance tokens to secure their responsibilities of evaluation and onboarding of new farmland.

LANDX CHOICE

LandX aims to create sustainable yield built upon regenerative finance including a commitment to reinvest 5% of protocol fees to sustainability programs via LandX Choice Fund:

- Education for land operators
- ESG programs, certification & research
- SDG goal certifications
- Financial inclusion training
- Agricultural monitoring technology
- Sustainable farming tools & equipment

Capital is used as a tool to invest in real-world yield opportunities and solve systemic issues and to regenerate natural environments.

The LandX team believes in providing financial flexibility to farmers, while creating new primitives in decentralized finance that offer access to a dependable asset class.

The LandX protocol is a bridge between real world assets and blockchain technology.

Landowners are able to enter agreements with investors on a global scale and receive capital for business expansion and long-term food security efforts.

LandX is creating a robust technological, legal and financial framework to connect farmers and investors in a brand new way.

“In the past it has been extremely difficult for farmers to raise capital and for crypto investors to have exposure to real revenue generating assets.

LandX provides an open, transparent, decentralized solution”

LandX.fi

To discuss partnerships, investments or for further information please contact
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